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Professor and Chair of the Department

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Professors

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Associate Professors

O. Alam, BSc (King Abdulaziz), MSc (Queen's), PhD (McGill)

B. G. Patrick, BCS (Concordia), MSc (Queen's), PhD, MBA (McGill)

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The study of computer science is central to our understanding of how the world processes, manages, and communicates information. It is a field which impacts all disciplines and has redefined how we work, interact, and conduct our day-to-day lives. The Department of Computer Science offers three streams of study: Computer Science (BSc program), Computing Systems (BA program), and Information Systems (joint-major BA/BSc program). Under the Computer Science stream, three specializations are offered: Theoretical Computer Science, Data Analytics, and Software Engineering.

Computer science specialists design, implement, and integrate software applications which may execute on any system of computing devices, from supercomputers to game boards. They use theories, methodologies, and techniques that help ensure that the system is robust and useable. Information systems specialists, on the other hand, are concerned with the flow, storage, and access to information both within and across organizational boundaries, encompassing systems that are both computational and non-computational. They require a broad systems perspective and an appreciation of both technical matters like databases, networks, and telecommunications, as well as non-technical matters such as ethics, law, and digital culture. The minor in Computer Studies offers the study of computing from a generalist point of view. This option may particularly appeal to students in the humanities, liberal arts, teacher education, or those with a general interest in computing.

Notes

- Students entering the Honours Bachelor of Science program in Computer Science are required to have completed a secondary school senior-level mathematics course in good standing.
- Courses in Computer Science fulfilling University science credit requirements for the BSc are marked "Sc" in the individual course descriptions.
- The same course may not simultaneously satisfy the requirements of both programs in a joint-major degree.
- Students may not graduate with a joint-major degree in any two of our three areas of study: Computer Science, Computing Systems, and Information Systems.

- Students majoring in Computer Science and who have had some programming experience may consider enrolling in both COIS 1010H and 1020H in the Fall term to increase their options for additional COIS courses earlier in their program.
- COIS 1010H and 1020H are offered in both the Fall and Winter terms.
- MATH 1110H and 1120H are recommended for students in a Computer Science program who are interested in postgraduate studies.
- For the program in Computer Science & Physics, see [p. 152](#).
- For the Postgraduate Certificate in Applied Artificial Intelligence, see [p. 582](#).

Bachelor of Science Program in Computer Science

The single-major Honours program. 20.0 credits including the following 12.5 credits:

- 4.0 COIS credits consisting of COIS 1010H, 1020H, 2020H, 2240H, 2300H, 3020H, 3380H, and 3400H
- 2.0 COIS credits at the 4000 level
- 2.5 COIS credits at the 3000 level or beyond in addition to the above
- 2.0 COIS credits in addition to the above
- 1.5 MATH credits consisting of MATH 1350H, 1550H, and 2600H
- 0.5 MATH credit from MATH 1005H or 1110H
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#)) and 14.0 science credits

The joint-major Honours program. 20.0 credits including the following 8.5 credits:

- 3.0 COIS credits consisting of COIS 1010H, 1020H, 2020H, 2240H, 2300H, and 3400H
- 1.0 COIS credit at the 4000 level
- 1.5 COIS credits at the 3000 level or beyond in addition to the above
- 1.0 COIS credit in addition to the above
- 1.5 MATH credits consisting of MATH 1350H, 1550H, and 2600H
- 0.5 MATH credit from MATH 1005H or 1110H
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#)) and 14.0 science credits

The single-major General program. 15.0 credits including the following 9.5 credits:

- 3.5 COIS credits consisting of COIS 1010H, 1020H, 2020H, 2240H, 2300H, 3020H, and 3400H
- 2.0 COIS credits at the 3000 level or beyond in addition to the above
- 2.0 COIS credits in addition to the above
- 1.5 MATH credits consisting of MATH 1350H, 1550H, and 2600H
- 0.5 MATH credit from MATH 1005H or 1110H
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#)) and 11.0 science credits

Bachelor of Arts Program in Computing Systems

The joint-major Honours program. 20.0 credits including the following 7.5 credits:

- 3.0 COIS credits consisting of COIS 1010H, 1020H, 1620H, 2240H, 2300H, and 3400H
- 0.5 COIS credit from COIS 2020H or 2620H
- 1.0 COIS credit at the 4000 level
- 1.5 COIS credits at the 3000 level or beyond in addition to the above
- 1.5 COIS credits in addition to the above
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#))

The single-major General program. 15.0 credits including the following 8.0 credits:

- 3.0 COIS credits consisting of COIS 1010H, 1020H, 1620H, 2240H, 2300H, and 3400H
- 0.5 COIS credit from COIS 2020H or 2620H
- 2.5 COIS credits at the 3000 level or beyond in addition to the above
- 2.0 COIS credits in addition to the above
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#))

Bachelor of Arts and Bachelor of Science Programs in Information Systems

The joint-major Honours program. 20.0 credits including the following 7.0 credits:

- 3.5 COIS credits consisting of COIS 1010H, 1020H, 1620H, 2240H, 2620H, 3030H, and 3850H
- 0.5 COIS credit at the 4000 level
- 1.5 COIS credits at the 3000 level or beyond in addition to the above
- 1.5 COIS credits in addition to the above
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#))
- For the BSc, 14.0 science credits are required including 1.0 MATH credit

The joint-major General program. 15.0 credits including the following 5.0 credits:

- 2.0 COIS credits consisting of COIS 1010H, 1020H, 1620H, and 2620H
- 2.0 COIS credits at the 3000 level or beyond
- 1.0 COIS credit in addition to the above
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#))
- For the BSc, 11.0 science credits are required including 1.0 MATH credit

Minors in Computing & Information Systems

Students who have fulfilled the requirements for a single-major or joint-major BA or BSc Honours degree in another subject may graduate with one of the following minors in computing or information systems.

The minor in Computer Studies consists of 5.0 COIS credits, at least 2.0 at the 3000 level or beyond.

The minor in Computer Science consists of the following 6.5 credits:

- 1.5 COIS credits consisting of COIS 1010H, 1020H, and 2020H
- 2.0 COIS credits at the 3000 level or beyond
- 1.5 COIS credits in addition to the above
- 1.0 MATH credit consisting of MATH 1350H and 1550H
- 0.5 MATH credit from MATH 1005H or 1110H

The minor in Data Analytics consists of the following 5.0 credits:

- 3.0 COIS credits consisting of COIS 1010H, 1020H, 1400H, 3400H, 3510H, and 4400H
- 0.5 COIS credit from COIS 2020H or 2620H
- 1.0 COIS credit at the 3000 level or beyond in addition to the above
- 0.5 COIS credit in addition to the above

The minor in Information Systems consists of the following 5.0 credits:

- 2.0 COIS credits consisting of COIS 1010H, 1020H, 1620H, and 2620H
- 2.0 COIS credits at the 3000 level or beyond
- 1.0 COIS credit in addition to the above

Specialization in Data Analytics

Program Coordinator

S. M. McConnell, BSc, MSc, PhD (Queen's)

Today, there is a clear trend away from single-core processors, even in contemporary desktop machines, and toward the analysis of large datasets in the realm of Big Data Science. The efficient analysis of large scientific and retail datasets will require a generation of programmers, analysts, and developers who are familiar with techniques suitable for the analysis of Big Data. The Specialization in Data Analytics provides a solid foundation in those techniques, including parallel programming, visualization, and predictive modelling.

Students who have fulfilled the requirements for a single-major Honours degree in Computer Science may graduate with a Specialization in Data Analytics if they have also successfully completed the following 4.0 credits:

- 4.0 COIS credits consisting of COIS 1400H, 3030H, 3510H, 3560H, 4350H, 4400H, 4470H, and 4550H

Specialization in Software Engineering

Program Coordinator

O. Alam, BSc (King Abdulaziz), MSc (Queen's), PhD (McGill)

The software engineer works as part of a team to design, implement, test, maintain, and document software solutions. Because these solutions are typically large and complex, the software engineer must deal on several levels of abstraction and draw on sound methodologies and practices to produce software which is correct, robust, maintainable, and extensible. The software engineer therefore bridges the gap between the requirements of a client and the software solution to meet those requirements. He or she must have not only technical skills, but also interpersonal skills to work within the dynamics of a group. The capstone of the Specialization in Software Engineering is a full-year team project (COIS 4000Y) which exercises all phases of software development on a real-world application.

Students who have fulfilled the requirements for a single-major Honours degree in Computer Science may graduate with a Specialization in Software Engineering if they have also successfully completed the following 9.5 credits:

- 4.5 COIS credits consisting of COIS 2420H, 3030H, 3040H, 3050H, 3320H, 3430H, 3850H, and 4000Y
- 1.0 credit from ADMN
- 1.0 arts credit from AGRS, CAST, CUST, ENGL, FREN, GESO, HIST, PHIL or SPAN
- 1.0 science credit from BIOL, CHEM, ERSC, FRSC, GEOG, or PHYS
- 2.0 additional credits outside of COIS and MATH

Specialization in Theoretical Computer Science

Program Coordinator

B. G. Patrick, BCS (Concordia), MSc (Queen's), PhD, MBA (McGill)

The theoretical computer scientist studies and develops software systems which interface most closely with the underlying hardware. These systems are grounded in both mathematical and non-mathematical paradigms such as object-orientation, relations, and complexity theory. Because the theoretical computer scientist bridges the gap between software and hardware systems, he or she develops applications which exploit the speed and power of digital computation, designing new models of computation and new methods of problem-solving. The Specialization in Theoretical Computer Science fully prepares students to pursue graduate studies in computer science.

Students who have fulfilled the requirements for a single-major Honours degree in Computer Science may graduate with a Specialization in Theoretical Computer Science if they have also successfully completed the following 8.0 credits:

- 3.0 COIS credits consisting of COIS 2320H, 3030H, 3050H, 3320H, 4050H, and 4100H
- 1.0 credit from ADMN
- 1.0 arts credit from AGRS, CAST, CUST, ENGL, FREN, GESO, HIST, PHIL, or SPAN
- 1.0 science credit from BIOL, CHEM, ERSC, FRSC, GEOG, or PHYS
- 2.0 additional credits outside of COIS and MATH

Computer Science Co-Op

Program Coordinator

B. Srivastava, BSc (Guelph), MSc (Trent)

Co-op is an opportunity for students to gain experiential learning which will help them get a head start on a career. The Computer Science Co-op assists graduating students in achieving a competitive edge while generating experience within the industry. Specifically, co-op gives the students a chance to obtain 12 months of practical experience on the job, develop or enhance technical skills, explore career options, network with a variety of individuals and companies, learn from mentors and tutors, enhance communication and presentation skills, and develop technical writing skills.

Co-op combines academic instruction and practical experience, allowing students to develop work skills and gain experience while obtaining a formal education. The work/study sequence in the Computer Science Co-op program consists of eight study terms interspersed with three paid work terms. Students are admitted on a full-time basis and complete the program over a four-year period.

Notes

- The co-op program is competitive with a limited number of spaces. Students who are not admitted directly may apply for admission after commencing studies at Trent and be admitted on a competitive basis provided there are available spaces in the program.
- Every effort is made by the department to find suitable work terms for students enrolled in the co-op, but the department cannot guarantee a work term opportunity for every student. The work term process is competitive and dependent upon many factors such as market conditions and the academic performance of a student. Co-op students secure positions through successful participation in the co-op interview process and/or their own job search activities. No more than two of the three work terms may be with the same employer (without permission).
- Students cannot enrol in more than one 0.5-credit course while on a work term without permission of the co-op program coordinator.
- Withdrawal from the co-op will not normally be approved while a student is on a work term.
- Students who are unable to obtain a suitable co-op opportunity in an upcoming work term are expected to continue with their academic program on a full-time basis. A missed work term can be made up, but in the event that a co-op student is unable to complete three work terms yet otherwise fulfills all degree requirements, the student will be awarded an Honours Bachelor of Science in Computer Science (and if applicable, any of the Specializations: Theoretical Computer Science, Data Analytics, and/or Software Engineering).
- Co-op students are responsible for paying tuition for their academic courses, ancillary fees, and levies. All Trent students are charged ancillary fees every Fall and Winter term. Co-op students will experience the same cycle for their ancillary fee payments regardless of when they are out on work terms. Co-op students do not pay tuition for their work terms. Co-op students are responsible for paying the compulsory Co-op Fee a total of five times during their degree program. The fee will be applied to their student account in Year 2 Fall, Year 2 Winter, Year 3 Fall, Year 3 Winter, and Year 4 Fall.
- Other than year 1, the schedule below is not mandatory, but any deviation could result in students requiring additional time to complete the program, as not all courses required for the degree are offered in every term.

	Fall	Winter	Summer
Year 1	Study Term 1	Study Term 2	Study Term 3
Year 2	Work Term 1	Study Term 4	Work Term 2
Year 3	Study Term 5	Study Term 6	Work Term 3
Year 4	Study Term 7	Study Term 8	–

Bachelor of Science Program in Computer Science – Co-op

- Students must successfully complete three non-credit work term courses (COIS 2700P, 3700P, and 4700P) prior to the final academic term.
- To remain in good standing in the co-op program, students must maintain a minimum overall average of 75% in their completed COIS courses. In addition, co-op students must continue to receive satisfactory evaluations during their work terms. Students who do not maintain these requirements will be required to meet with the co-op program coordinator and will be considered for continuation on a case-by-case basis.
- To be eligible for a first co-op work term, students must successfully complete COIS 1400H, 2020H, 2240H, 2430H, 3400H, and 3430H. Students who do not meet this requirement but have already arranged a work term, will not be enrolled in the co-op work term course, and the work term cannot be counted towards the co-op program.
- Students should follow the suggested course schedule on the Co-op page of the department website (trentu.ca/cois) for the Year 1 study terms (Fall, Winter, and Summer). Recommended courses change frequently based on market needs.
- Students returning from a work term must submit a report describing the nature of the work undertaken and the experience gained. A grade of Pass (P) or Fail (F) is assigned based on the report and the employer's assessment. The report is normally due within the first two weeks of the study term following the completion of a work term.

The single-major Honours program. 20.0 credits including the following 12.5 credits:

- 5.5 COIS credits consisting of COIS 1010H, 1020H, 1400H, 2020H, 2240H, 2300H, 2430H, 3020H, 3380H, 3400H, and 3430H
- 2.0 COIS credits at the 4000 level
- 2.0 COIS credits at the 3000 level or beyond in addition to the above
- 1.0 COIS credit in addition to the above
- 1.5 MATH credits consisting of MATH 1350H, 1550H, and 2600H
- 0.5 MATH credit from MATH 1005H or 1110H
- In addition to the program requirements listed above, students must satisfy the University degree requirements (see [p. 15](#)), including 0.5 credit from the Approved Indigenous Course List (see [p. 19](#)) and 14.0 science credits

Please consult the academic timetable for information on courses that will be offered in 2024–2025, including when they will be scheduled.

» COIS-MDST 1010H: The Digital World (Sc)

Digital systems have redefined how we work, communicate, and play—just think about the World Wide Web, mobile camera phones, video games, and e-business. Core topics examine the underlying technologies of both computing and information systems and how they have become an integral and indispensable part of our daily lives.

» COIS 1020H: Programming for Computer Science (Sc)

Programming for computer science requires a solid foundation in the software development process, including algorithmic design, abstraction, implementation, testing, and documentation. Core topics include sequencing, selection, iteration, simple data types, expressions, and arrays, as well as the object-oriented notions of classes, methods, inheritance, and polymorphism. Pre- or co-requisite: COIS 1010H or PHYS 1001H or MATH 1110H. Students without prior programming experience should take COIS 1010H as a prerequisite.

» **COIS 1400H: Introduction to Data Science (Sc)**

Develops a solid foundation in the main concepts of data science and programming in Python. Core topics include repetition and selection structures, algorithm design techniques, file types, big data, data mining, and data visualization. Prerequisite: COIS 1020H.

» **COIS-ADMN 1620H: Introduction to Information Systems (Sc)**

Wherever and whenever information is required, an information system is also required. Core topics include the use of information systems for strategic advantage, their basic underlying technologies, the types of information systems and how they are constructed, managed, and replaced, as well as their ethical and legal use. Prerequisite: COIS 1010H.

» **COIS 2020H: Data Structures and Algorithm (Sc)**

Data structures define how data are represented. Algorithms define how data structures are manipulated. Using more advanced programming language features, core topics include arrays, linked lists, hash tables, binary heaps, and binary search trees. Prerequisite: COIS 1020H. Recommended: MATH 1350H.

» **COIS 2040H: Object Oriented Programming (Sc)**

The C++ and/or Java languages are utilized to examine object-oriented concepts. These concepts can include classes, objects, inheritance, encapsulation, polymorphism, class deviation, abstract classes, interfaces, static class members, object construction and destruction, exceptional handling, function overloading and overriding, overload resolution, container classes, and/or template classes. Prerequisite: COIS 1020H.

» **COIS 2240H: Software Design and Modelling (Sc)**

Good software design and modelling is a necessary prerequisite for the production of software which is correct, robust, and maintainable. Using the standard Unified Modeling Language (UML) to specify design, core topics include use cases; classes and class membership; aggregation, composition, and inheritance; virtual functions and polymorphism; state diagrams; and design patterns. Prerequisite: COIS 1020H (or both COIS 1520H and COIS-ADMN 2620H).

» **COIS-PHYS 2250H: Electronics (Sc) (see Physics & Astronomy)**

» **COIS 2300H: Computer Organization (Sc)**

Computer organization is the level of abstraction between the digital logic and the operating system. At this level, computing subsystems refer to hardware pieces built from lower-level building blocks. Core topics include the central processing unit, memory hierarchy, and input/output organization. Topics are supplemented with assembly language programming. Prerequisite: COIS 1020H (or 1520H).

» **COIS 2320H: Digital Logic (Sc)**

Digital logic describes how computer hardware actually works at the logic gate and circuit level. Core topics include Boolean algebra, Karnaugh maps, the minimization of Boolean functions and the design of combinational and sequential circuits, including adders, decoders, multiplexers, flip-flops, and memory circuits. Prerequisite: COIS 1020H (or 1520H).

» **COIS 2400H: R in Data Science (Sc)**

An introduction to the programming language R and its applications in data science. A wide range of topics are covered, with emphasis on techniques for data gathering, cleaning and manipulation, text and numerical analysis, and visualization to provide relevant applications in data science. Prerequisite: COIS 1400H.

» **COIS-PHIL 2410H: Symbolic Logic (Sc) (see Philosophy)**

» **COIS 2430H: Web Development I: Front-End (Sc)**

Front-end web development requires a strong understanding of the technologies used on the client side of a web application. Topics include front-end languages HTML, CSS, and JavaScript, as well as important concepts like responsive design, accessibility, usability, version control, and regular expressions. Prerequisite: COIS 1020H or 1400H. Not open to students with credit for COIS 3420H.

» **COIS 2520H: Professionalism and Ethics in Computing (Sc)**

Identifies and explores some of the fundamental social, legal, and ethical issues inherent in the discipline of computing. Topics include social context, analytical tools, professional ethics, intellectual property, privacy and civil liberties, professional communication, and equity issues. Prerequisite: COIS 1020H.

» **COIS-MATH 2600H: Discrete I: Structures (Sc) (see Mathematics)**

» **COIS-ADMN 2620H: Systems Analysis and Design (Sc)**

Systems analysis and design identifies the informational needs/resources of an organization and specifies appropriate information systems to meet the needs and exploit the resources for strategic advantage. Core topics include the current methodologies, techniques, and tools used for both the analysis and design. Managerial and organizational issues are also discussed. Prerequisite: COIS-ADMN 1620H.

» **COIS-FRSC 2750H: Computer Crime and Forensics (Sc)**

Computer crime is the fastest-growing area of illegal activity in the world. Users beware! After some background information (how computers work, number systems, information-hiding algorithms), we examine the schemes and techniques used by computer criminals, the forensic techniques used to catch the criminals, and ways to prevent victimization. Recommended prerequisite: COIS 1010H.

» **COIS-MDST 2800H: Digital Culture**

Interactive digital technology and the convergence of computing, communications, public media, and entertainment have had profound cultural implications. Core topics address current concerns such as the World Wide Web, multimedia design, virtual spaces, social domains, and the blogosphere. Recommended prerequisite: COIS 1010H. This course cannot be counted as a science credit.

» **COIS-MDST 2830H: Multimedia and Design (Sc)**

Multimedia concepts, such as photo and video editing, visual effects, 3D assets, and even writing style serve as tools to explore concepts and challenges in modern user experience design, with a focus on web pages, game design, and user interfaces. Principles of design covers core concepts such as the discoverability of features, understanding diverse human users, complexity of feedback, and how computer systems interpret and react to human actions. Recommended prerequisite: COIS 1010H.

» **COIS 3020H: Data Structures and Algorithms II (Sc)**

A continuation of COIS 2020H, this course introduces more complex data structures and algorithmic techniques. Core topics include adjacency matrices and lists, randomly built binary search trees, augmented data structures, tries, binomial heaps, B-trees, disjoint sets, and quad trees. Prerequisite: COIS 2020H. Recommended: COIS-MATH 2600H.

» **COIS 3030H: Software Specification and Development (Sc)**

The development of large software systems relies on sound methodologies and techniques to manage the inherent complexity and to produce quality software which satisfies the client or customer. Core topics include various software development paradigms and their usage, as well as the importance of feasibility studies, requirements analysis, and specification. Prerequisite: COIS 2240H.

» **COIS 3040H: Advanced Software Architecture and Design (Sc)**

Software architecture is concerned with the high-level design, interface, and relationships among the components that make up a software system. Testing ensures that these components work properly on their own and within an integrated system. Core topics include architecture description languages, as well as maintenance strategies and various testing methodologies. Prerequisite: COIS 2020H and 3030H.

» **COIS 3050H: Formal Languages and Automata (Sc)**

Formal languages and automata are the underlying theoretical foundations of computer science. Core topics include regular languages and finite automata, context-free languages and pushdown automata, the Turing machine and its languages. Real world applications, unsolvable problems, and their relevance to the semantics of programming are also discussed. Prerequisite: COIS 1020H. Recommended: COIS-MATH 2600H.

» **COIS 3060H: Software Processes (Sc)**

Introduces students to software processes, project management practices, and software design methodologies such as waterfall, iterative and agile approaches. Taken in preparation for Software Engineering co-op work terms. Prerequisite: 5.0 COIS credits including COIS 2040H.

» **COIS 3065H: Cross Platform Application Development (Sc)**

This course provides students insight—both applied and theoretical—into the field of cross-platform application development. Emphasis is placed on the qualities of good interface design and the processes, tools, and frameworks required to develop applications for current and emerging mobile and desktop environments. Prerequisite: COIS 2040H and 2430H.

» **COIS-MATH 3180H: Numerical and Computational Methods (Sc) (see Mathematics)**

» **COIS-PHYS 3200Y: Electricity and Magnetism (Sc) (see Physics & Astronomy)**

» **COIS-PHIL 3270H: Philosophy of Artificial Intelligence (see Philosophy)**

» **COIS 3320H: Fundamentals of Operating Systems (Sc)**

An operating system is the software component of a computing system which is responsible for the management and accessibility of the hardware resources. Core topics include process management, synchronization, processor allocation, primary and secondary storage management, file management, security, and user interfaces. Topics are illustrated using contemporary operating systems. Prerequisite: COIS 2020H and 2300H.

» **COIS-MATH 3350H: Linear Programming (Sc) (see Mathematics)**

» **COIS-PHIL-MDST 3370H: Cyberethics (see Philosophy)**

» **COIS 3380H: Systems Programming in C (Sc)**

Systems programming produces software which provides services to the computer hardware. Using systems programming techniques in a Unix-style environment and using machine-oriented programming languages such as C/C++, core topics include procedural programming, shell programming, pipes, file processing, system calls, signals, and basic network programming. Prerequisite: COIS 2020H or 2300H.

» **COIS 3400H: Database Management Systems (Sc)**

A database is a structured collection of data, whereas a database management system facilitates the creation, querying, and maintenance of databases. Core topics include the relational model, entity-relationship diagrams, relational algebra, SQL, database application development, storage and indexing, and normalization. Topics are illustrated using a relational DBMS. Prerequisite: COIS 1020H (or both COIS 1520H and COIS-ADMN 2620H).

» **COIS 3430H: Web Development II: Server-side and Frameworks (Sc)**

Following the foundations set in COIS 2430H, this course encourages the exploration of more advanced topics in web applications, including design patterns, server-side programming, database integration, APIs, security, and front-end frameworks, along with development processes like build tools, testing and CI/CD. Prerequisite: COIS 2430H or 3420H. Pre- or co-requisite: COIS 3400H.

» **COIS 3510H: Data Visualization (Sc)**

Data visualization is a main step in the analysis of data in a wide range of scientific research areas as well as business applications. We discuss general approaches, tools, and techniques for the visualization of various types of data, including spatial data, graph data, and time series data. Prerequisite: 10.0 university credits or permission of instructor. Recommended: COIS 1020H (or 1520H).

» **COIS-CUST-MDST 3533H: Game Studies (see Cultural Studies)**

» **COIS 3550H: Applied Artificial Intelligence and Machine Learning (Sc)**

Fundamental artificial intelligence (AI) and machine learning concepts, algorithms, and techniques. The focus is on why and what can be done, along with available tools and applications in solving real-world problems, rather than the finer details of how it could be built. The course uses TensorFlow with provided Python code. Prerequisite: COIS 1020H.

» **COIS-MDST 3560H: Human-Computer Interaction (Sc)**

The study of human-computer interaction incorporates ideas from computer science, behavioural science, and design to improve interaction between people and computers. Core topics include multidisciplinary, intuitive design, computer-supported co-operative education/work, ubiquitous computing, prototyping, collaboration, social interaction, augmented reality, haptics, heuristics, interfaces, workstation independence. Prerequisite: 5.0 university credits.

» **COIS-ANTH-MDST 3745H: Virtual Worlds: The Anthropology of Online Communities (see Anthropology)**

» **COIS-ADMN 3750H: Principles of Online Marketing (Sc)**

Businesses today must effectively leverage the Internet and develop an online strategy which complements their traditional business practices. Core topics include website best practices, online marketing and customer relationships, positioning businesses online, and social networking for small and large enterprises. Topics are consolidated with a professional business proposal. Prerequisite: COIS-ADMN 1620H or ADMN 1000H or permission of instructor.

» **COIS-MDST 3820H: History and Impact of Computing (Sc)**

The history of digital computation is relatively recent, but all around us. By introducing the key people whose insight, inventiveness, and industry have defined the digital world, a framework is developed within which all students can appreciate the fundamental milestones of computing and their impact on the world. Prerequisite: 5.0 university credits.

» **COIS-ADMN 3850H: Fundamentals of Project Management (Sc)**

Good project management focuses on the practical realities of getting things done with limited time and resources according to specifications determined by others—with successful outcomes. This course prepares students to write the professional certification exams offered by the Project Management Institute and the Project Management Association of Canada. Prerequisite: 10.0 university credits or COIS 2240H or COIS-ADMN 2620H or permission of instructor.

» **COIS 3860H: Internship Project (Sc)**

Students work at a site external to the department for 80–120 hours. In recognition of the scope of possible projects, this work can either be performed during 2–3 one-week (not necessarily consecutive) sessions, or during 10–15 full-day sessions in consecutive weeks. Students meet on a regular basis with their faculty mentor. Prerequisite: A minimum 75% cumulative average and 10.0 university credits including 5.0 COIS credits.

» **COIS 3901H: Reading Course (Sc)**

Students are required to submit a course proposal and have it approved by the departmental curriculum committee before enrolling.

» **COIS 4000Y: Software Engineering Project (Sc)**

The software engineering project is a capstone course and provides students with the practical experience in the analysis, design, implementation, testing, and documentation of a real-world software product. Working as a team with clients, the project draws substantially on the theoretical knowledge gained in all previous COIS courses. Prerequisite: COIS 3040H. Pre- or co-requisite: COIS-ADMN 3850H. Recommended: COIS 3400H and 3430H. Not open to students with credit for COIS 4850H.

» **COIS 4010H: Current Topics in Data Science (Sc)**

Provides an overview of the current status of data science in a variety of focus fields and guides students in the preparation for their COIS 4020H project. Students discuss and present current research, evaluation methods, data set characteristics, and tools from their chosen areas. Prerequisite: 10.0 university credits including two of COIS 2020H, 3510H, or 4400H, or permission of the instructor.

» **COIS 4020H: Data Science Project (Sc)**

This course guides students in the implementation of their capstone data science project. Under faculty supervision, students propose, develop, implement, and evaluate a data science solution. Prerequisite: COIS 4010H. Open only to students in the Data Science program.

» **COIS 4050H: Advanced Algorithms (Sc)**

Algorithmic techniques provide generalized approaches to solving a wide variety of problems. Algorithmic analysis provides the basis of comparison among different solutions. Core topics include the algorithmic techniques of divide-and-conquer, dynamic programming, greedy algorithms, approximation algorithms, and string matching as well as the analytical techniques of problem classification, asymptotic analysis, amortized analysis, and recurrence relations. Prerequisite: COIS 2020H and COIS-MATH 2600H. Recommended: COIS 3020H.

» **COIS 4060H: Applied Software Engineering (Sc)**

Presents an overview of applications and active research in software engineering. Introduces students to software engineering processes in new and emerging fields not covered by more traditional software development focused courses. May include topics such as real-time and embedded software, cloud computing, Dev Ops, and signal and image processing. Prerequisite: COIS 2240H.

» **COIS 4065H: Software Quality (Sc)**

Introduces systematic testing of software systems. Software verification, reviews, metrics, quality assurance, and prediction of software reliability and availability. Related management issues. Focuses on software quality and testing. Topics may include white and black box testing, test coverage, integration testing, system testing, model-based testing, and testing tools. Testing within the context of different software development methodologies is discussed. Prerequisite: COIS 2040H and 2240H.

» **COIS 4100H: Comparative Programming Languages (Sc)**

Evaluating and selecting the most appropriate programming language for an application is a critical step in software development. Core topics include the types of programming languages as well as the principal design and implementation issues for control structures, data types, subprograms, exception handling, and concurrency. Prerequisite: COIS 2020H and 1.0 COIS credit at the 3000 or 4000 level.

» **COIS 4310H: Computer Networks (Sc)**

A computer network links together two or more computers to enable the transfer of data. Core topics include data communication, network hardware and software, network architecture and protocols, local area and wide area networks, end-to-end routing protocols, wireless networks, Quality of Service, multimedia support, and network management. Prerequisite: COIS 3320H.

» **COIS 4350H: High Performance Computing (Sc)**

High performance computing is the use of advanced computer architectures to solve problems which require significant processing power, memory access, or storage. Core topics include advanced computer architectures, programming for shared and distributed memory machines, networking issues, caching, performance evaluation, and parallel algorithms. Topics are supplemented with case studies. Prerequisite: COIS 3320H or 3380H.

» **COIS 4370H: Computer and Information Security (Sc)**

Security of data and information systems is a vital component of an organization's infrastructure. Core topics include the key concepts and legal implications of information security, risk and security management, asset security, and principles and policies for securing data. Attention is given to the methods of confirming and monitoring compliance to agreed-upon policies. Prerequisite: COIS 3320H or 3380H.

» **COIS 4400H: Data Mining (Sc)**

Data mining is the extraction of previously unknown and potentially useful patterns from large data sets. Core topics include the statistical foundations of data mining, sampling mechanisms, missing data, clustering, classification, and outlier detection. Case studies cover a wide variety of applications such as astronomy and marketing. Prerequisite: 1.0 COIS credit at the 3000 or 4000 level and one of COIS 2020H or COIS-ADMN 2620H.

» **COIS 4450H: Big Data (Sc)**

Big data applications are encompassing commercial and scientific, moving towards exascale analytics. Students from a variety of disciplines are introduced to the challenges and opportunities in this field, with the goal of providing them with theoretical and hands-on experience in the area of big data analytics. Prerequisite: COIS 3380H and 3510H.

» **COIS 4470H: Modelling and Simulation (Sc)**

Modelling and simulation enables us to construct mathematical or computer-based representations of real-world phenomena. Three steps are typically followed: modelling the key parameters, implementing a program to simulate behaviour, and analyzing the results. Core topics include pseudo-random number generation, discrete-event and next-event simulation, output analysis, and queuing theory. Prerequisite: MATH 1550H and COIS 2020H.

» **COIS 4480H: Computer Graphics (Sc)**

Computer graphics are everywhere, from computer-animated movies and games to virtual reality and telepresence. This course introduces the fundamental mathematical and computational concepts of computer graphics. Topics include hardware, the OpenGL suite, line drawing, two-dimensional transformations, two-dimensional clipping algorithms, three-dimensional representations, three-dimensional clipping, line removal techniques, splines, and ray tracing. Prerequisite: COIS 2020H and MATH 1350H. Recommended: MATH 1110H.

» **COIS 4550H: Artificial Intelligence (Sc)**

Artificial intelligence is the study of those techniques which create perceptions of "machine intelligence" and "intelligent agents." Topics may include but are not limited to expert systems, various evolutionary learning systems such as genetic algorithms, genetic programming, and neural networks. The impact and ethics of artificial intelligence are also examined. Prerequisite: 10.0 university credits including COIS 2020H.

» COIS 4850H: Information Systems Project (Sc)

A capstone course which provides students with the practical experience in the analysis and design (but not implementation) of a real-world information system. Working as a team with clients, the project draws substantially on the theoretical knowledge gained in all previous COIS courses. Prerequisite: COIS-ADMN 2620H, COIS 3030H, and COIS 3850H. Recommended: COIS 3400H. Not open to students with credit for COIS 4000Y.

» COIS 4900Y, 4901H, 4902H: Advanced Reading Course (Sc)

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